

TABLA DE DERIVADAS			
TIPO	FUNCIÓN	DERIVADA	EJEMPLOS
Potencial	$y = x^a$ $y = f^a$	$y' = ax^{a-1}$ $y' = af^{a-1} \cdot f'$	$f(x) = (2x+5)^4$ $f'(x) = 4(2x+5)^3 \cdot 2 = 8(2x+5)^3$
Raíz cuadrada	$y = \sqrt{x}$	$y' = \frac{1}{2\sqrt{x}}$	$f(x) = \sqrt{x^2 - 3x}$
	$y = \sqrt{f}$	$y' = \frac{f'}{2\sqrt{f}}$	$f'(x) = \frac{2x-3}{2\sqrt{x^2 - 3x}}$
Exponencial	$y = e^x$	$y' = e^x$	$f(x) = e^{4x-3}$
	$y = e^f$	$y' = f' \cdot e^f$	$f'(x) = 4 \cdot e^{4x-3}$
	$y = a^x$	$y' = a^x \cdot \log a$	$g(x) = 5^{x^2+x}$
	$y = a^f$	$y' = f' \cdot a^f \cdot \log a$	$g'(x) = (2x+1) \cdot 5^{x^2+x} \cdot \log 2$
Logaritmo neperiano	$y = \log x$	$y' = \frac{1}{x}$	$f(x) = \log(2x^3 + 5x)$
	$y = \log f$	$y' = \frac{f'}{f}$	$f'(x) = \frac{6x^2 + 5}{2x^3 + 5x}$
Seno	$y = \operatorname{sen} x$	$y' = \cos x$	$f(x) = \operatorname{sen}(x^2 + x)$
	$y = \operatorname{sen} f$	$y' = f' \cdot \cos f$	$f'(x) = (2x+1) \cdot \cos(x^2 + x)$
Coseno	$y = \cos x$	$y' = -\operatorname{sen} x$	$f(x) = \cos 5x$
	$y = \cos f$	$y' = f'(-\operatorname{sen} f)$	$f'(x) = 5 \cdot (-\operatorname{sen} 5x) = -5\operatorname{sen} 5x$
Tangente	$y = \operatorname{tg} x$	$y' = \frac{1}{\cos^2 x} = 1 + \operatorname{tg}^2 x$	$f(x) = \operatorname{tg} 5x$
	$y = \operatorname{tg} f$	$y' = f' \cdot \frac{1}{\cos^2 f} = f' \cdot (1 + \operatorname{tg}^2 f)$	$f'(x) = 5 \cdot \frac{1}{\operatorname{tg}^2 5x} = \frac{5}{\operatorname{tg}^2 5x}$ O bien $f'(x) = 5 \cdot (1 + \operatorname{tg}^2 5x)$
Cotangente	$y = \operatorname{ctg} x$	$y' = \frac{-1}{\operatorname{sen}^2 x}$	$f(x) = \operatorname{ctg} e^{3x}$
	$y = \operatorname{ctg} f$	$y' = f' \cdot \frac{-1}{\operatorname{sen}^2 f}$	$f'(x) = 3e^{3x} \cdot \frac{-1}{\operatorname{sen}^2 e^{3x}} = \frac{-3e^{3x}}{\operatorname{sen}^2 e^{3x}}$

DERIVADAS DE OPERACIONES CON FUNCIONES		
Suma o resta	$(f \pm g)' = f' \pm g'$	$f(x) = 5x^2 + e^{3x} - \operatorname{sen} x$ $f'(x) = 10x + 3e^{3x} - \cos x$
Producto	$(f \cdot g)' = f' \cdot g + g' \cdot f$	$f(x) = e^x \operatorname{sen} x$ $f'(x) = e^x \cdot \operatorname{sen} x + \cos x \cdot e^x = e^x (\operatorname{sen} x + \cos x)$
Cociente	$\left(\frac{f}{g}\right)' = \frac{f' \cdot g - g' \cdot f}{g^2}$	$f(x) = \frac{2x+1}{2x-1}$ $f'(x) = \frac{2(2x-1) - 2(2x+1)}{(2x-1)^2} = \frac{4x-2-4x-2}{(2x-1)^2} = \frac{-4}{(2x-1)^2}$